

P100 Unit Diagnostic Report

Disk Degradation - SMART Media Integrity Errors

Node (planename)	N070AV
Airline	AVA (Avianca)
Serial Number	W20041875122
Disk Model	ADATA IM2P33E8-001TD (NVMe, 1.02 TB)
Disk Serial	2M042L16E4XD
Diagnostic Date	August 20, 2026
Diagnosis	Disk degradation - elevated SMART media integrity errors
Current Operational State	Disk is currently mounted and operational; degradation confirmed via SMART

Diagnostic Summary

Unit N070AV boots normally and all filesystems (boot, root, kvmroot, wwwdata partitions) mount cleanly with no errors reported at the OS/EXT4 level. However, a SMART health check of the internal NVMe disk reveals a significant and abnormal count of media and data integrity errors: **126,992,617** - more than 2,000 times higher than the baseline observed on a comparable healthy unit (N098AV, ~53,000 errors). While the disk's overall SMART self-assessment still reports 'PASSED' and available spare capacity remains at 100%, this volume of integrity errors indicates active, significant degradation of the storage media that is not yet visible through normal filesystem operation but represents meaningful risk of future failure - including the kind of failure that has previously caused units to be returned from modem repair with a newly-surfaced disk problem.

Diagnostic Steps Performed

1. Verified the unit boots normally into its OS with all partitions correctly mounted (boot/EFI, root, secondary root, /kvmroot, /mnt/wwwdata) and no EXT4 mount errors in kernel logs.
2. Installed smartmontools (not present by default on this unit) to enable NVMe SMART diagnostics.
3. Ran a full SMART health report on the internal NVMe disk (smartctl -a /dev/nvme0n1).
4. Compared key SMART attributes against a known-healthy baseline unit to assess severity: Media and Data Integrity Errors, Unsafe Shutdowns, Percentage Used, Available Spare, and operating Temperature.

Key SMART Findings

Attribute	N070AV (this unit)	Healthy baseline (N098AV)	Assessment
Media and Data Integrity Errors	126,992,617	~53,368	Severely elevated
Unsafe Shutdowns	1,852	~772	Elevated
Percentage Used	12%	~7%	Slightly elevated, not critical alone

Available Spare	100%	100%	Normal
Temperature	67°C	38°C	Elevated, close to 75°C warning threshold
Overall SMART self-assessment	PASSED	PASSED	Does not reflect the integrity error volume

Conclusion: despite the disk currently being operational and passing its own overall self-assessment, the media integrity error count is severe enough to indicate active degradation of the NVMe storage media, elevated well beyond a healthy comparison unit. Combined with an elevated unsafe shutdown count and operating temperature close to the warning threshold, this unit is at meaningful risk of surfacing a disk-related failure in the near term - including after other repairs (e.g. modem work) are completed and the unit is returned to service. Preventive disk replacement is recommended.

Appendix: Technical Evidence (logs and command output)

Unit Identification

N070AV
AVA
Serial Number: W20041875122
UUID: 03000200-0400-0500-0006-000700080009
Wake-up Type: Power Switch
SKU Number: SGP100-713020-00;M0;W0;B0-15
Date: Thu Aug 20 14:29:49 UTC 2026

Block Devices and Mounted Partitions (lsblk -l)

```
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
nvme0n1 259:0 0 953.9G 0 disk
nvme0n1p1 259:1 0 512M 0 part /boot/efi
nvme0n1p2 259:2 0 28G 0 part /
nvme0n1p3 259:3 0 28G 0 part
nvme0n1p4 259:4 0 39.1G 0 part /kvmroot
nvme0n1p5 259:5 0 858.4G 0 part /mnt/wwwdata
```

Kernel Messages - NVMe and Filesystem Mounts (dmesg)

```
[ 5.421239] nvme nvme0: pci function 0000:02:00.0
[ 5.647256] nvme nvme0: 8/0/0 default/read/poll queues
[ 5.654717] nvme0n1: p1 p2 p3 p4 p5
[ 8.488029] EXT4-fs (nvme0n1p2): mounted filesystem with ordered data mode. Opts: (null)
[ 10.012127] EXT4-fs (nvme0n1p2): re-mounted. Opts: (null)
[ 11.950169] EXT4-fs (nvme0n1p4): mounted filesystem with ordered data mode. Opts: (null)
[ 13.215279] EXT4-fs (nvme0n1p5): mounted filesystem with ordered data mode. Opts: (null)
```

No filesystem errors present - all mounts completed cleanly.

NVMe Device Info (nvme list)

```
Node SN Model Namespace Usage Format FW Rev
-----
/dev/nvme0n1 2M042L16E4XD ADATA_IM2P33E8-001TD 1 567.24 GB / 1.02 TB 512 B + 0 B TS0305H
```

Full SMART Health Report (smartctl -a /dev/nvme0n1)

Model Number: ADATA_IM2P33E8-001TD
Serial Number: 2M042L16E4XD
Firmware Version: TS0305H
Namespace 1 Size/Capacity: 1,024,209,543,168 [1.02 TB]
Namespace 1 Utilization: 567,187,861,504 [567 GB]
Warning Comp. Temp. Threshold: 75 Celsius
Critical Comp. Temp. Threshold: 80 Celsius

SMART overall-health self-assessment test result: PASSED

SMART/Health Information (NVMe Log 0x02)
Critical Warning: 0x00

Temperature: 67 Celsius
Available Spare: 100%
Available Spare Threshold: 10%
Percentage Used: 12%
Data Units Read: 33,388,199 [17.0 TB]
Data Units Written: 69,641,840 [35.6 TB]
Host Read Commands: 704,291,796
Host Write Commands: 1,471,994,444
Controller Busy Time: 62,197
Power Cycles: 2,101
Power On Hours: 24,351
Unsafe Shutdowns: 1,852
Media and Data Integrity Errors: 126,992,617
Error Information Log Entries: 0
Warning Comp. Temperature Time: 2
Critical Comp. Temperature Time: 0
Thermal Temp. 1 Transition Count: 87
Thermal Temp. 1 Total Time: 46874

Error Information (NVMe Log 0x01, max 256 entries)
No Errors Logged